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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, MAY 2024
Second Semester

21MAB102T – ADVANCED CALCULUS AND COMPLEX ANALYSIS
(For the candidates admitted during the academic year 2021-2022, 2022-2023 & 2023-2024)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer **ALL** Questions

	Marks	BL	CO	PO

1. $\int_0^1 \int_0^x dxdy$ is equal to — 1 2 1 1

(A) 1 (B) 1/2
(C) 2 (D) 3

2. Area of the double integral in polar co-ordinates is equal to — 1 1 1 2

(A) $\int_R \int drd\theta$ (B) $\int_R \int r^2 drd\theta$
(C) $\int_R \int (r+1) drd\theta$ (D) $\int_R \int rdrd\theta$

3. In polar, the integral value of $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dxdy =$ — 1 1 1 2

(A) $\int_0^{\pi/2} \int_0^\infty e^{-r^2} drd\theta$ (B) $\int_0^{\pi/4} \int_0^\infty e^{-r} drd\theta$
(C) $\int_0^{\pi/2} \int_0^\infty e^{-r^2} rdrd\theta$ (D) $\int_0^{\pi} \int_0^\infty e^{-r^2} drd\theta$

4. $\int_0^1 \int_0^2 \int_0^4 dxdydz$ — 1 2 1 1

(A) 8 (B) 4
(C) 2 (D) 6

5. The maximum directional derivative of $\phi = xyz^2$ at (1,0,3) is — 1 1 2 1

(A) 9 (B) $9\bar{j} + 6\bar{k}$
(C) $2\bar{i} + 4\bar{j} - \bar{k}$ (D) –9

6. Curl (grad ϕ) — 1 2 2 2

(A) –1 (B) 1
(C) 0 (D) 2

7. The value of 'a' so that $\vec{F}=(x+z)\vec{i}+(3x+ay)\vec{j}+(x-5z)\vec{k}$ is solenoidal 1 2 2 2

(A) 1 (B) 4
(C) 2 (D) 3

8. The value of $\int_C \vec{F}.d\vec{r}$ is 1 1 2 1

(A) $\iiint_V div\vec{F}dv$ (B) $\iint_S div\vec{F}.\hat{n}ds$
(C) $\iint_S \vec{F}.\hat{n}ds$ (D) $\iint_S curl\vec{F}.\hat{n}ds$

9. L[sinh a t] is 1 1 3 1

(A) $\frac{a}{s^2-a^2}$ *if* $s>|a|$ (B) $\frac{s}{s^2-a^2}$ *if* $s>|a|$
(C) $\frac{s}{s^2+a^2}$ *if* $s>|a|$ (D) $\frac{a}{s^2+a^2}$ *if* $s>|a|$

10. An example of a function for which the Laplace transform does not exist is 1 2 3 2

(A) $f(t)=t^2$ (B) $f(t)=\tan t$
(C) $f(t)=\sin t$ (D) $f(t)=e^{-at}$

11. Find $L^{-1}\left[\frac{1}{(s+a)^2}\right]$ 1 1 3 2

(A) te^{at} (B) te^{-at}
(C) t (D) e^{at}

12. Using the initial value theorem, find the value of the function $f(t)=ae^{-bt}$ 1 1 3 1

(A) a (B) a^2
(C) ab (D) 0

13. If u and v are harmonic then u+iv is 1 1 4 1

(A) Harmonic (B) Need not be analytic
(C) Analytic (D) Continuous

14. The invariant points of the transformation $w=\frac{1-iz}{z-i}$ 1 1 4 1

(A) 0 (B) ±i
(C) ±2 (D) ±1

15. The transformation w = z+c where c is a complex constant represents 1 1 4 2

(A) Rotation (B) Magnification
(C) Translation (D) Magnification and rotation

16. An analytic function with constant modulus is 1 1 4 2

(A) Zero (B) Analytic
(C) Constant (D) Harmonic

17. The value $\int_C \frac{zdz}{z-3}$ where C is the circle |z|=1 is — 1 2 5 2

(A) 0
(B) $\pi/2$ i
(C) $\pi/2$
(D) 3

18. If f(z) is analytic inside an on C, the value of $\int_C \frac{f(z)dz}{z-a}$, where C is the simple closed curve and 'a' is any point within 'C' is — 1 2 5 2

(A) $f(a)$
(B) $2\pi i f(a)$
(C) $\pi i f(a)$
(D) 0

19. Let C:|z–a|=r be a circle, then f(z) can be expanded as a Taylor's series if — 1 1 5 1

(A) f(z) is a defined function with in C
(B) f(z) is a analytic function within C
(C) f(z) is not a analytic function within C
(D) f(z) is analytic function outside C

20. If $f(z)=\frac{\sin z - z}{z^3}$ then — 1 1 5 2

(A) z=0 is a simple pole
(B) z=0 is a pole of order 2
(C) z=0 is a removable singularity
(D) z=0 is a zero f f(z)

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO

21. a. Change the order of integration in $\int_0^b \int_0^{\frac{a}{b}(b-y)} xydxdy$ and then evaluate it. — 8 4 1 2

(OR)

b. Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{(x+y+z)} dzdydx$. — 8 4 1 2

22. a.i. Find the directional derivative of the function $\phi = 2yz + z^2$ in the direction of the vector $\vec{i} + 2\vec{j} + 2\vec{k}$ at the point (1,–1,3). — 4 3 2 2

ii. Find the angle between the normals to the surface $xy = z^2$ at the points (1,4,2) and (–3,–3,3). — 4 2 2 2

(OR)

b. Verify Green's theorem in a plane for $\int_C \left[\left(3x^2 - 8y^2\right)dx + \left(4y - 6xy\right)dy\right]$, where C is the boundary of the region defined by the lines x=0, y=0 and x+y=1. — 8 4 2 2

23. a. Find $L^{-1}\left[\frac{s^2}{\left(s^2+a^2\right)\left(s^2+b^2\right)}\right]$ using convolution theorem. — 8 4 3 2

(OR)

	b.	Using Laplace transform solve $y''+4y'+5y=0$ given $y(0)=0, y'(0)=1$.	8	4	3	2
24.	a.	Determine the analytic function $f(z)=u+iv$ such that $u-v=e^{x}(\cos y-\sin y)$.	8	4	4	1
		(OR)				
	b.	Find the bilinear transformation which maps the points 1, i, −1 of z-plane onto the points 0, 1, ∞ of w-plane.	8	3	4	2
25.	a.	Evaluate $\int_{C}\frac{(z-1)dz}{(z+1)^{2}(z-2)}$ where C is $\lvert z-i\rvert=2$.	8	4	5	1
		(OR)				
	b.	Expand $f(z)=\frac{1}{z^{2}-3z+2}$ in a Laurent's series in the region (i) $1<\lvert z\rvert<2$ (ii) $0<\lvert z-1\rvert<2$.	8	3	5	2

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Questions

			Marks	BL	CO	PO
26.		Verify Gauss divergence theorem for $\vec{F}=4xz\vec{i}-y^{2}\vec{j}+yz\vec{k}$ over the cube bounded by x=0, x=1, y=0, y=1, z=0, z=1.	15	4	2	2
27.		Evaluate $\int_{0}^{2\pi}\frac{d\theta}{2+\cos\theta}$ using contour integration.	15	4	5	2

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